VARUN BAJPAI

Full-Stack Developer | MERN | AI/RAG | CSE Undergraduate

Meerut, UP | varunbajpai16@gmail.com | Available for Internship | LinkedIn | GitHub | LeetCode

SUMMARY — 3rd-year CSE student expected to graduate in 2027, building and deploying MERN full-stack applications and RAG/agentic AI projects. Seeking a software development or full-stack internship to contribute to APIs, database-backed applications, and AI-powered workflows.

EDUCATION

B.Tech, Computer Science & Engineering — Meerut Institute of Engineering and Technology, Meerut | 3rd Year | CGPA: 7.4/10 | Expected Graduation: 2027

TECHNICAL SKILLS

Languages: C++, JavaScript, Python, SQL

Frontend: React.js, Redux Toolkit, Tailwind CSS, HTML5, CSS3, Responsive Design

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Express Middleware

Databases: MongoDB, PostgreSQL, Vector Databases

AI: LangChain, LangGraph, RAG, Agentic AI

Tools & Platforms: Git, GitHub, Postman, Vercel, Render, Cloudinary

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

PROJECTS

TutorMate — MERN Stack | Live Demo | GitHub

• Developed and deployed a tutor discovery platform serving 100+ users, with Google OAuth and role-based access for teachers, students, and parents.

• Implemented MongoDB 2dsphere geospatial queries for proximity-based teacher discovery and designed 10+ REST APIs with Express.js and Mongoose.

• Built a responsive React/Tailwind interface; deployed the frontend on Vercel and backend on Render.

YouTube Clone — MERN Stack | Live Demo | GitHub

• Built a video-sharing application with upload, playback, likes, comments, and subscriptions; tested with 50+ users.

• Designed MongoDB schemas and 15+ REST APIs; implemented JWT authentication with refresh tokens and Cloudinary media delivery.

NexusAI — AI / RAG | Live Demo | GitHub

• Built and deployed a retrieval-augmented generation application using LangChain and LangGraph to retrieve relevant context from a vector database and generate contextual responses.

• Organized the workflow around retrieval and response generation to provide context-aware answers.

EcoTrust — Environmental Intelligence Platform | Live Demo | GitHub

• Built a platform ingesting simulated CEMS sensor data to monitor industrial emissions, compliance, and factory-level alerts.

• Implemented validation and trust scoring using data completeness, consistency, sensor reliability, temporal stability, and validation history.

• Developed a role-based React dashboard with geospatial factory monitoring and LangChain/LangGraph workflows, backed by Node.js, Express, and MongoDB.

LEADERSHIP & ACTIVITIES

• **Smart India Hackathon (SIH) — Team Lead:** Led a 6-member team developing EcoTrust, an environmental intelligence solution for industrial emissions monitoring; coordinated task distribution, sprint planning, technical execution, and final presentation. The project was eliminated in Round 3.

• **Self-Initiated Projects:** Independently architected, built, and deployed MERN applications using Vercel and Render.

ACHIEVEMENT

• Solved 200+ LeetCode problems across arrays, trees, graphs, and dynamic programming.

PROFILES & PROJECT LINKS

LinkedIn | GitHub | LeetCode | TutorMate | YouTube Clone | NexusAI | EcoTrust | Portfolio